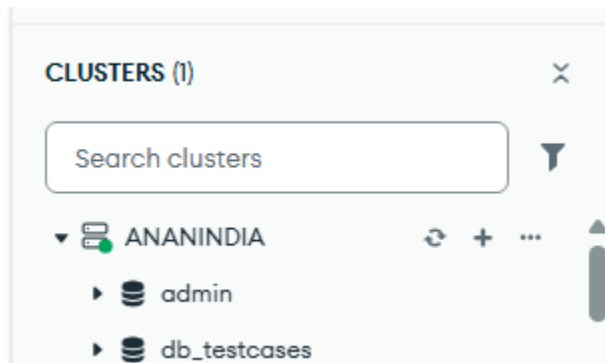


RAG INGESTION THROUGH BACKEND-TEST CASES

LETS TRY TO INGEST THE TEST CASES THROUGH BACKEND TO THE DATABASE

Lets connect to Data Explorer and connect to the cluster we have created

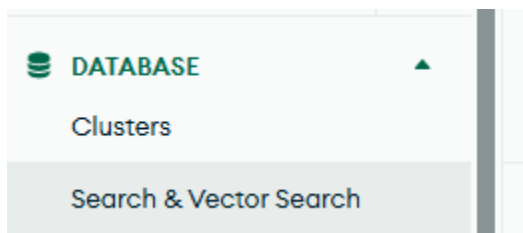
Then lets create our DB. Here its named as db_testcases



Inside that lets create our collection . here its named as “test_cases”



LETS GOAHEAD AND CREATE THE SEARCH INDEX. LETS GO TO THE FOLLOWING NAVIGATION



LETS CREATE A NEW SEARCH INDEX AND SELECT THE FOLLOWING

Atlas Search

Full-text search for relevance-based app features.

Vector Search

For semantic search and AI applications.

How do you want to set up your vector data?

[Learn more about different data setup options](#)

Automated Embedding

NEW

START FREE

MongoDB generates and manages embeddings for your text fields – no external models or APIs needed.

Bring your own embeddings

Use vectors you've already generated or plan to generate with your own

Index Name

Database and Collection

▼ db_testcases

● test_cases

Configuration Method

Select how you would like to configure your index. You can also define indexes through the [Atlas Admin API](#), [Atlas CLI](#), [MongoDB Shell](#), or supported [Drivers](#).

Visual Editor

Define index definitions in a more guided experience.

JSON Editor

Edit the raw index definition with an embedded JSON editor.

Cancel

Next

Copy paste all the data in this place to this screen

```
src > config > {} testcases-vector-index.json > [] fields >  
1 {  
2   "fields": [  
3     {  
4       "type": "vector",  
5       "path": "embedding",  
6       "numDimensions": 1024,  
7       "similarity": "cosine"  
8     },  
9     {  
10      "type": "filter",  
11      "path": "id"  
12    },  
13  ]  
}
```

JSON Editor

[View Atlas Vector Search Docs](#)

Search Index Type: vectorSearch

```
3 {  
4   "type": "vector",  
5   "path": "<field-name>",  
6   "numDimensions": 0,  
7   "similarity": "<similarity-function>"  
8 }  
9 ]  
10 }
```

Click next and confirm the index creation. Below is the screenshot of the index I already have created

db_testcases	test_cases	vector_index_test_cases	READY View status details	✓	vectorSearch	"embedding", "id", "module", "title", "description" +? more
--------------	------------	-------------------------	------------------------------	---	--------------	---

When we search for the collection currently we will have 0 data

On the project setup shared lets open it in VS code and navigate to the env file

Update the mongo DB URI for the DB we have created and pass the collectionname and the vector_index name we have created to import the test case. In my case they are the follows

```
MONGODB_URI="mongodb+srv://iamanandhiram_db_user:iamanandhiram@ananindia.ohf5xrt.mongodb.net/?appName=ANANINDIA"
#MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cLuster0.c5L4wjg.mongodb.net/?retryWrites=true&w=majority&appName=ANANINDIA
DB_NAME="db_testcases"
#DB_NAME="RagTest_cases"

COLLECTION_NAME="test_cases"
VECTOR_INDEX_NAME="vector_index_test_cases"
BM25_INDEX_NAME="bm25_search"

# Mistral AI Configuration (for embeddings)
MISTRAL_API_KEY=DyWzoDAEbiKwLc9At1VU4gonk00CDXpg
```

THE NEXT STEP IS TO CONVERT THE TESTCASES IN EXCEL TO JSON. THIS CAN BE ACHIEVED BY BELOW STEP

```
PS D:\Users\a.natarajan\OneDrive - Corpay\personal\rag-mongo-demo-backend\rag-mongo-demo-backend> node src/scripts/data-conversion/excel-to-json.js

PS D:\Users\a.natarajan\OneDrive - Corpay\personal\rag-mongo-demo-backend\rag-mongo-demo-backend>
✓ Converted 6354 rows from "testcases" into src/data/testcases.json
```

LETS CONVERT THIS JSON DATA TO EMBEDDINGS AND PUSH TO OUR COLLECTION USING THE INDEX WE HAVE CREATED

```
PS D:\Users\A.natarajan\OneDrive - Corpay\personal\rag-mongo-demo-backend\rag-mongo-demo-backend> node src/scripts/embeddings/create-embeddings-batch-mistral.js
```

BATCH PROCESS HAS COMMENCED. THIS IS DUE TO TH SETTINGS WE HAVE MADE EXCLUSIVELY ON CODE TO PROCESS ONLY 50 TEST CASES PER RUN

```
[Batch 2/128] Processing 50 testcases with Mistral AI...
[Batch 3/128] Processing 50 testcases with Mistral AI...
[Batch 2/128] Success! Cost: $0.000818 | Tokens: 8175
[Batch 3/128] Success! Cost: $0.000768 | Tokens: 7682
[Batch 1/128] Success! Cost: $0.000736 | Tokens: 7355
Waiting 1000ms before next batch group...
```

ONCE THIS IS COMPLETED WE CAN VIEW ALL THE TEST CASES LOADED INTO THE COLLECTION WE HAVE MENTIONED

Documents 4.4K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ✚ Explain Reset Find Options ▾

25 1 - 25 of 4350

```
{
  "_id": ObjectId('6a105bc21fbc4f5c85c1ff59'),
  "module": "Registration",
  "id": "TC_005"
}
```

```
Success Rate: 100.0%
Total Cost: $0.117741
Total Tokens: 1,177,411
Average Cost per Test Case: $0.00001853
Average Tokens per Test Case: 185
Cost Savings: ~90% cheaper than OpenAI embeddings
```

Type a query: { field: 'value' } or [Generate query](#) ✚ Explain Reset Find Options ▾

25 1 - 25 of 6354

```
{
  "_id": ObjectId('6a105bc21fbc4f5c85c1ff59'),
  "module": "Registration",
  "id": "TC_005"
}
```

INGESTION THROUGH UI – USER STORIES

FOLLOW SIMILAR STEPS OF CREATING A NEW COLLECTION AND VECTOR INDEX TO INGEST THE USERSTORIES

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'Data Explorer' sidebar shows a tree view with 'db_testcases' expanded, containing 'User_Story' and 'test_cases'. The main panel displays the 'User_Story' collection under the 'db_testcases' database. It shows a 'Documents' tab with 0 documents, and an 'Indexes' tab with 1 index. The index is named 'vector_index_user_story' and is of type 'vectorSearch'. The index fields are listed as 'embedding', 'projectName', 'statusCategory', 'priority', 'risk', and 'more'. The status is 'READY'.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields
db_testcases	User_Story	vector_index_user_story	READY View status details	✓	vectorSearch	"embedding", "projectName", "statusCategory", "priority", "risk", ... more

NOW LETS OPEN THE PROJECT FOLDER IN VS CODE AND MAKE CHANGES TO THE ENV FILE TO BE IN SYNC WITH OUR VECTOR db COMMECTION, INDEX NAME AND THE COLLECTION NAME

```
1 MONGODB_URI="mongodb+srv://iamanandhram_db_user:iamanandhram@ananindia.ohf5
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c5l4wjq.mongodb.net/?
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6
7 COLLECTION_NAME="test_cases"
8 VECTOR_INDEX_NAME="vector_index_test_cases"
9 BM25_INDEX_NAME="bm25_search"
10
11
12 USER_STORIES_COLLECTION_NAME="User_Story"
13 USER_STORIES_VECTOR_INDEX_NAME="vector_index_user_story"
14
15
16 #USER_STORIES_COLLECTION_NAME="user_stories"
17 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
18
19
20 # Mistral AI Configuration (for embeddings)
21 MISTRAL_API_KEY=DyWzoDAEbiKwLc9At1VU4gonk00CDXpg
22 MISTRAL_EMBEDDING_MODEL="mistral-embed"
```

LETS RUN THE COMMANDA TO START THE FRONT END AND BACKEND SIMULTAMEOUSLY

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Users\A.natarajan\OneDrive - Corpay\personal\rag-mongo-demo-ui\rag-mongo-demo-ui> npm run dev

[1] > rag-mongo-demo@1.0.0 client
[1] > cd client && npm start
[1]
[1]
[1] > client@0.1.0 start
[1] > react-scripts start
[1]
[0] 🚀 Server running on port 3001
[0] 🚀 API available at http://localhost:3001/api
```

THE UI HAS SUCCESSFULLY OPENED. NOW FIRST CONVERT THE USERSTORIES TO JSON AND THEN WE CAN UPLOAD THEM

Name	Status	Date modified	Type	Size
 testcases.xlsx		5/14/2026 8:02 AM	Microsoft Excel W...	918 KB
 testcases_500.xlsx		5/14/2026 9:56 AM	Microsoft Excel W...	104 KB
 userstories.xlsx		5/14/2026 8:02 AM	Microsoft Excel W...	26 KB

<

RAG Pipeline

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

SELECT EXCEL FILE (USER STORIES)

userstories.xlsx

25.6 KB

Selected

Sheet Name

stories

Name of the Excel sheet to convert (default: stories)

CONVERT USER STORIES

RESET

Description

Status

Priority

Assignee

Reporter

Story Points

Components

Labels

Sprint

Epic

User Stories converted successfully!

NOW TO CONVERT THE JSON TO EMBEDDINGS, SELET THE NEXT OPTION, CHOOS THE JSON FILE FROM UI AND PROCEED TO EMBEDDING CONVERSION, WE CAN SEE OUR COLLECTION UPDATED WITH THE DOCUMENTS

Embeddings & Store

Create and manage embeddings

Settings

Configure environment

Available Files

12 JSON files found

REFRESH

	File Name	Size	Modified ↓	Type
☒	stories-1779461555277.json	198.29 KB	5/22/2026 8:22:35 PM	JSON
	stories-1778735419879.json			

Processing Embeddings...

0/1

Current file: user-stories

Progress: 0%

Job ID: job-1779461643798-5y...

⚡ This process continues even if you refresh the page!

Actions

Process selected files

PROCESSING...

Batch processing started! Job ID: job-1779461643798-5y6f5svo7

Starting batch processing for 1 files...

Actions

Process selected files

CREATE EMBEDDINGS

Debug Info:
Selected Files: 1 (stories-1779461555277.json)
Processing: false
Button Disabled: false
Files loaded: 12

TEST SELECTION

SELECT FIRST FILE

Processing Time: ~10-30 seconds per file

Successfully processed all 1 files!

Batch processing started! Job ID: job-1779461643798-5y6f5svo7

Starting batch processing for 1 files...

CLUSTERS (1)

User_Story

test_cases

local

sample mfix

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

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```
_id: ObjectId('6a106e0e3e00013230fc16b3')
key: "HC-257"
summary: "Consultant Dashboard My Inpatients"
```